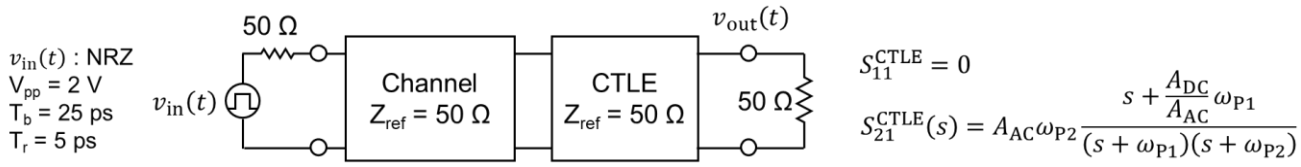
Homework: CTLE simulation and design

The s2p of a channel is provided to you. Your task is to design a CTLE to compensate the lowpass effect of the channel. The input source $v_{\text{in}}(t)$ is a 40 Gbps NRZ signal with risetime 5 ps. Assume impedance matching, so that the transfer function from $V_{\text{in}}(f)$ to $V_{\text{out}}(f)$ is simply $H = S_{21}^{\text{Ch}} S_{21}^{\text{CTLE}}/2$.



In this homework, we use the informative transfer function of USB 3 shown above. The parameters are ω_{P1}, ω_{P2}, A_{DC}, and A_{AC}. These are the parameters that you need to optimize (tune). Constraints:

- Pole location: $1\text{ GHz} \le f_{\text{P1}}, f_{\text{P2}} \le 40\text{ GHz}$
- Maximum gain: $\left|S_{21}^{\text{CTLE}}\right| \le 6\text{ dB}$
- DC attenuation: $-20\text{ dB} \le A_{\text{DC}} \le +6\text{ dB}$

Design goal: after equalization, center EH ≥ 300 mV.

Design (tuning) procedure:

1. First you need to complete the Python code correctly.
2. Choose a set of parameters ω_{P1}, ω_{P2}, A_{DC}, and A_{AC}.
3. Compute the overall transfer function $H = S_{21}^{\text{Ch}} S_{21}^{\text{CTLE}}/2$.
4. Let the input be N-bit random NRZ. Perform sectional convolution to obtain $v_{\text{out}}(t)$.
5. Construct the statistical eye diagram and observe the EH.
6. If EH target is not met, repeat steps 2~5 with a different set of parameters.

After the CTLE optimization is done, provide the following items in your report:

1. The magnitude plot (dB) of your optimized S_{21}^{CTLE}, and the parameters you chose.
2. The post-channel eye diagram and the post-CTLE eye diagram.
3. Three snapshots of the post-CTLE waveform, at different timing, each with width 1 ns, in comparison with the post-channel (un-equalized) waveform.

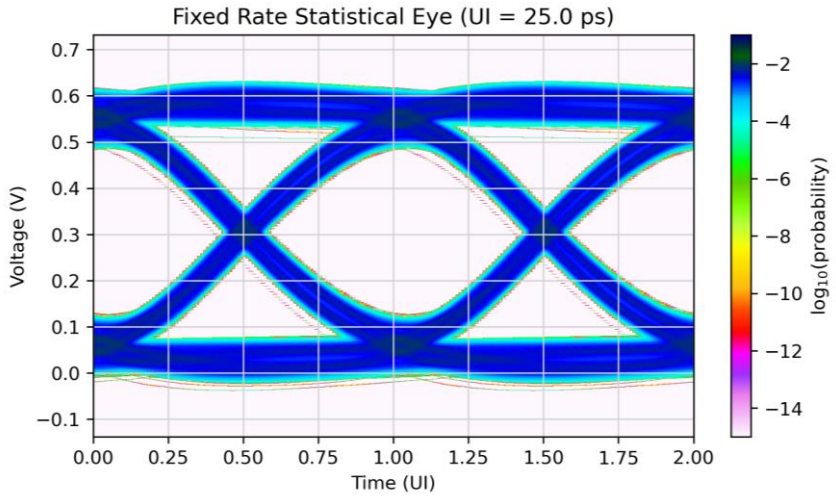
Optional problems:

Implement your CTLE in ADS ChannelSim to test its performance. See appendix.

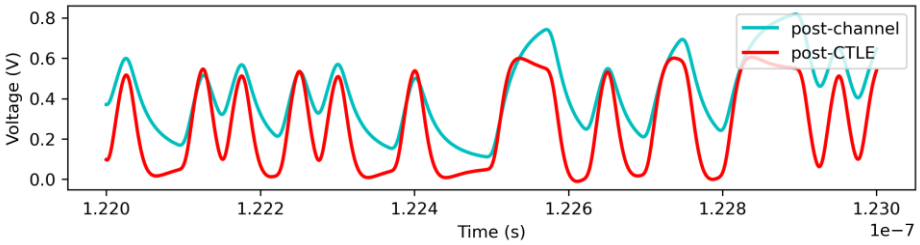
Note: In this homework, the Python code for doing sectional convolution and building stateye are packaged into functions. In addition, the stateye code is different (faster) than in previous homework. Also, in generating random bits, a seed value is used to guarantee generating the same pattern each time, thus providing comparable results.

Appendix 1: Example answer

Post-CTLE eye:

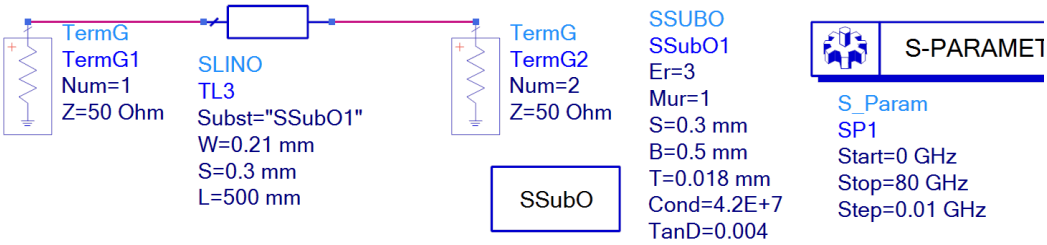


Waveform comparison

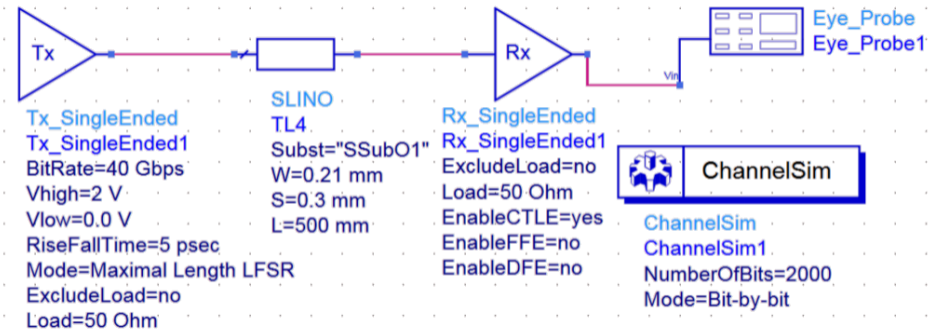


Appendix 2: ADS setting

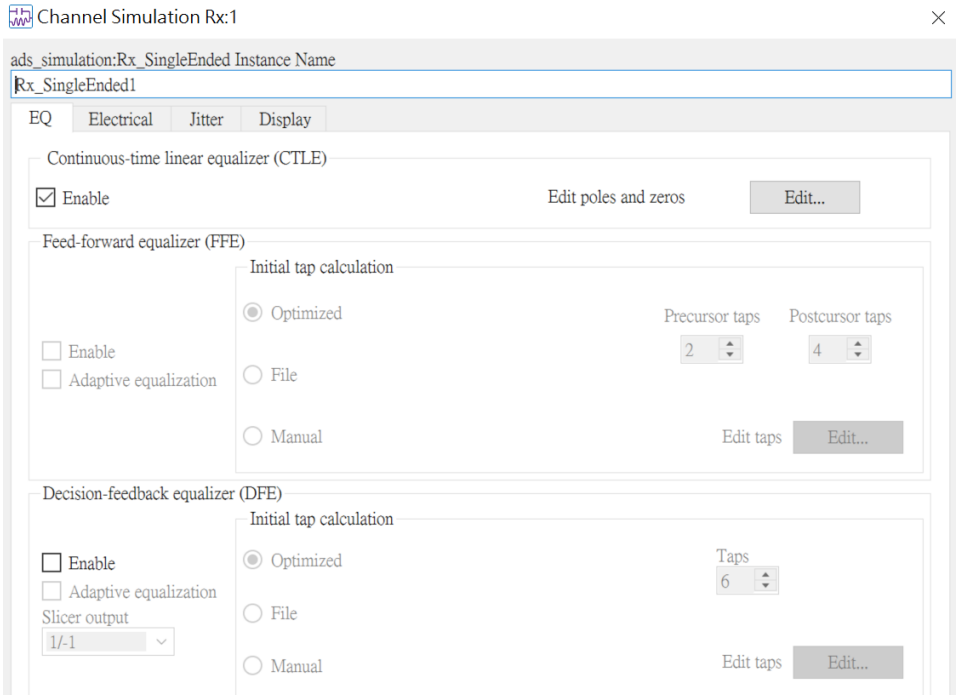
The channel in this homework is a 50-cm strip line. It is simulated in ADS as follows:



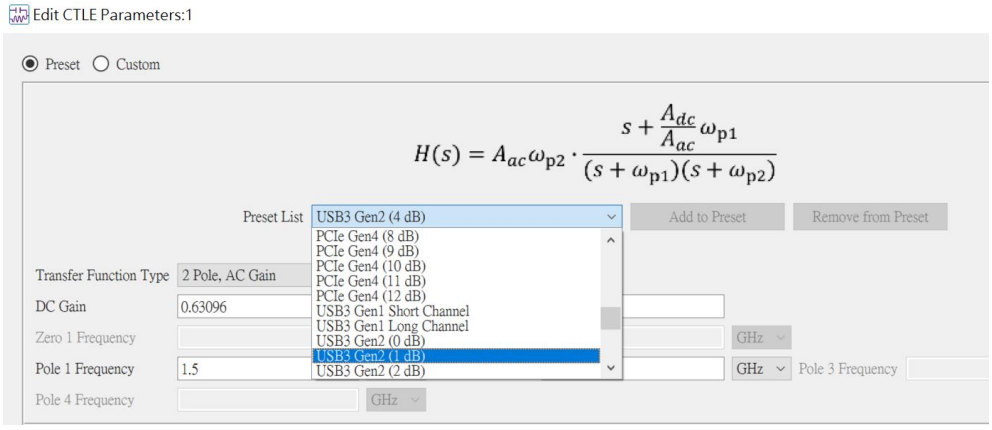
To simulate the eye diagram, use the ChannelSim tool:



In the “Rx” component, enable the CTLE:



In the CTLE setting, choose the USB template, and enter your optimized parameters



When plotting, choose the “density” to see the eye diagram

